Example LP4: A company produces two type of concrete:

Type A contains: 30% cement, 40% gravel, and 30% sand.

Type B contains: 10% cement, 30% gravel, and 60% sand.

Type A costs \$10 per kilogram, and type B costs \$2 per kilogram.

Objective: *How many kilograms of each type of concrete should we buy to minimize costs ? The concrete that we buy altogether should contain at least 3kg of cement, 3kg of gravel, and 6kg of sand.*

Sol: Let x_1 =amount of cement type A that we buy

x_2 =amount of cement type B that we buy

The cost will be: $f(x_1, x_2) = 10x_1 + 2x_2$

The amount of cement will be: $0.3x_1 + 0.1x_2 \geq 3$

The amount of gravel will be: $0.4x_1 + 0.3x_2 \geq 3$

The amount of sand will be: $0.3x_1 + 0.6x_2 \geq 6$

$x_1 \geq 0, \quad x_2 \geq 0$

Multiply by 10

We obtain an LP problem:

"Objective function" or "Cost function"

Minimize $f(x_1, x_2) = 10x_1 + 2x_2$

subject to
$$\begin{cases} 3x_1 + x_2 \geq 30 \\ 4x_1 + 3x_2 \geq 30 \\ 3x_1 + 6x_2 \geq 60 \\ x_1, x_2 \geq 0 \end{cases}$$

In matrix notation:

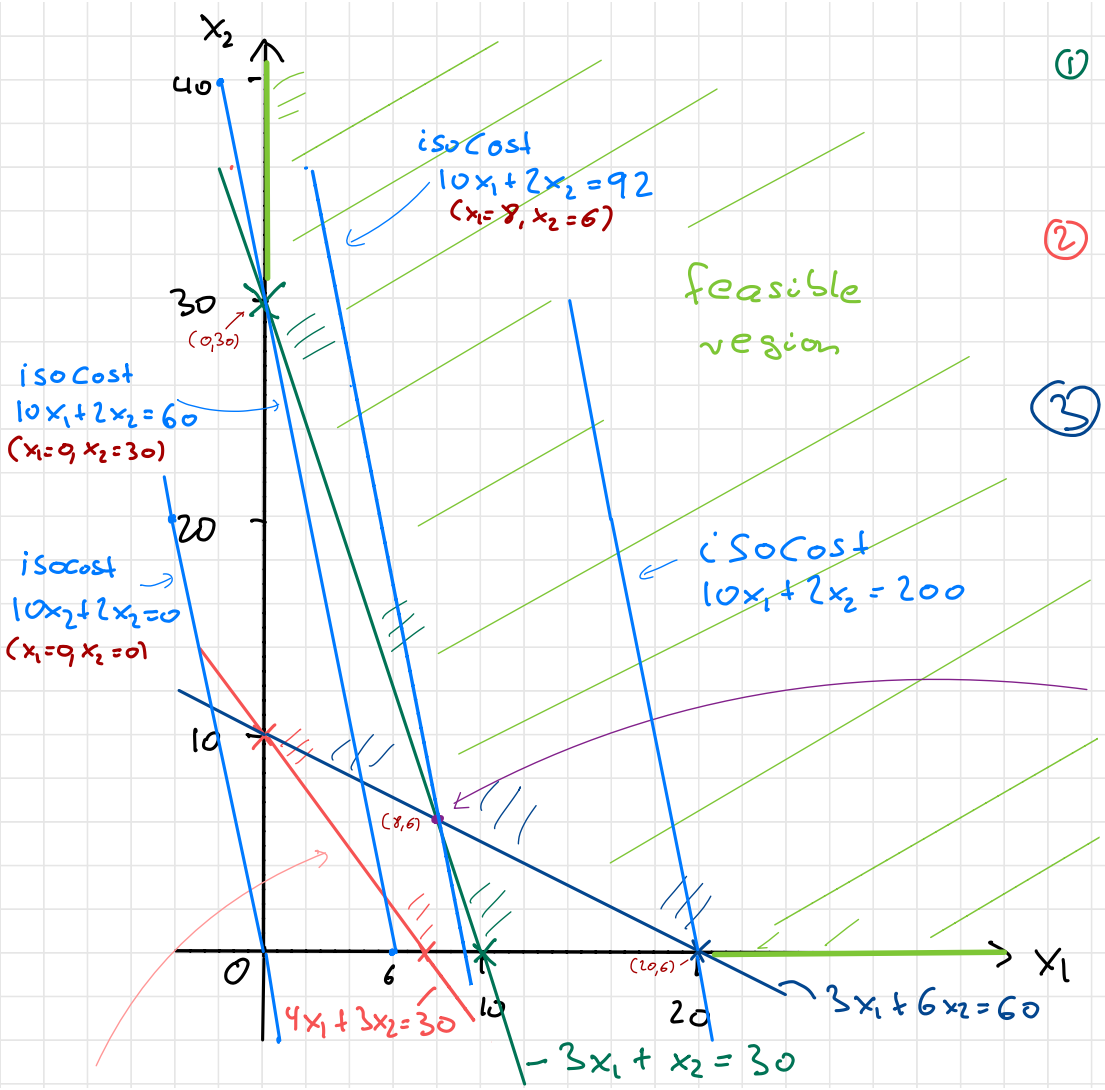
$$\text{minimize} \quad f(\mathbf{x}) = \mathbf{c}^T \mathbf{x}$$

$$\text{Subject to} \quad A\mathbf{x} \geq \mathbf{b}$$

$$\mathbf{x} \geq \mathbf{0}$$

where $\mathbf{c}^T = [\,10, 2\,]$, $A = \begin{bmatrix} 3 & 1 \\ 4 & 3 \\ 3 & 6 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 30 \\ 30 \\ 60 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Sketch the region Ω satisfying the constraints



① $3x_1 + x_2 = 30$ line

x_1-intercept $(x_2=0)$: $x_1=10$

x_2-intercept $(x_1=0)$: $x_2=30$

② $4x_1 + 3x_2 = 30$ line

x_1-intercept $(x_2=0)$: $x_1=7.5$

x_2-intercept $(x_1=0)$: $x_2=10$

③ $3x_1 + 6x_2 = 60$ line

x_1-intercept $(x_2=0)$: $x_1=20$

x_2-intercept $(x_1=0)$: $x_2=10$

find this point of intersection:

$$3x_1 + 6x_2 = 60 \quad (1)$$
$$3x_1 + x_2 = 30 \quad (2)$$

$(1)-(2)$: $5x_2 = 30 \Rightarrow x_2 = 6$

By (1), $x_1 + 2x_2 = 20$

$x_1 = 20 - 2x_2 = 20 - 2\cdot 6 = 8$

Point $(8,6)$

① We notice that the line $4x_1 + 3x_2 = 30$ is <u>not</u> part of the boundary, and therefore the condition

$$4x_1 + 3x_2 \geq 30$$

can be removed

② The feasible region is unbounded !!
We can still <u>minimize</u> the cost function though.
[But the max. does <u>not</u> exist]

③ The <u>parallel</u> lines $10x_1 + 2x_2 = k$ are called <u>isocost</u> lines. We want to find the <u>smallest</u> possible k, so that the line touches the feasible region.

(4) The feasible region has 3 "corners".

$(20,0)$, $(8,6)$, $(0,30)$

(5) Sketch isocost lines.

$$10x_1 + 2x_2 = k \qquad \left[x_2 = \frac{k}{2} - 5x_1 \right]$$

x_2-intercept $\frac{k}{2}$ slope -5

<u>Method 1:</u>

<u>we notice</u>: if we shift these lines to the <u>left</u> the value of k <u>decreases</u>. From the sketch, k is smallest at the "corner" $(0,30)$ of the feasible region.

<u>Answer:</u> $x_1 = 0$, $x_2 = 30$, Cost $= 10 \cdot 0 + 2 \cdot 30 = \underline{\underline{60}}$ (Min. Cost)

Method 2: Plug the 3 "Corners" into the cost function $f(x_1, x_2) = 10x_1 + 2x_2$.

$(20,0)$: $f(20,0) = 200$

$(8,6)$: $f(8,6) = 92$

$(0,3)$: $f(0,3) = 60 \leftarrow$ Min!

How to graphically solve an LP-problem in 2-dim.

1) Sketch the feasible region [using the constraints]

2) If the feasible region is <u>bounded</u>, then by the Extreme Value Theorem, the <u>max</u> and <u>min</u> of the objective function exist

If it is <u>unbounded</u>, the max or min may exist, or may <u>not</u> exist.

3) The feasible region is a polygon.

The max/min [if they exist] will occur at the "corners" of the feasible region.

Exercise 1: Write the LP-problem in vector form, and find its solution graphically:

maximize $\quad f(x_1, x_2) = x_1 + x_2$

subject to

$$x_2 - x_1 \le 1 \quad \rightarrow \text{sketch } x_2 - x_1 = 1 \quad (1)$$

$$x_1 + 6x_2 \le 15 \quad \rightarrow \text{sketch } x_1 + 6x_2 = 15 \quad (2)$$

$$4x_1 - x_2 \le 10 \quad \rightarrow \text{sketch } 4x_1 - x_2 = 10 \quad (3)$$

$$x_1, x_2 \ge 0$$

Find points of intersection:

(1) and (2)

$-x_1 + x_2 = 1 \quad (1)$

$x_1 + 6x_2 = 15 \quad (2)$

(1) + (2): $7x_2 = 16$

$x_2 = \frac{16}{7}$

$\underset{(1)}{\Rightarrow} x_1 = x_2 - 1 = \frac{16}{7} - 1 = \frac{9}{7}$

$\left(\frac{9}{7}, \frac{16}{7}\right)$

(2) and (3):

$x_1 + 6x_2 = 15 \quad (2)$

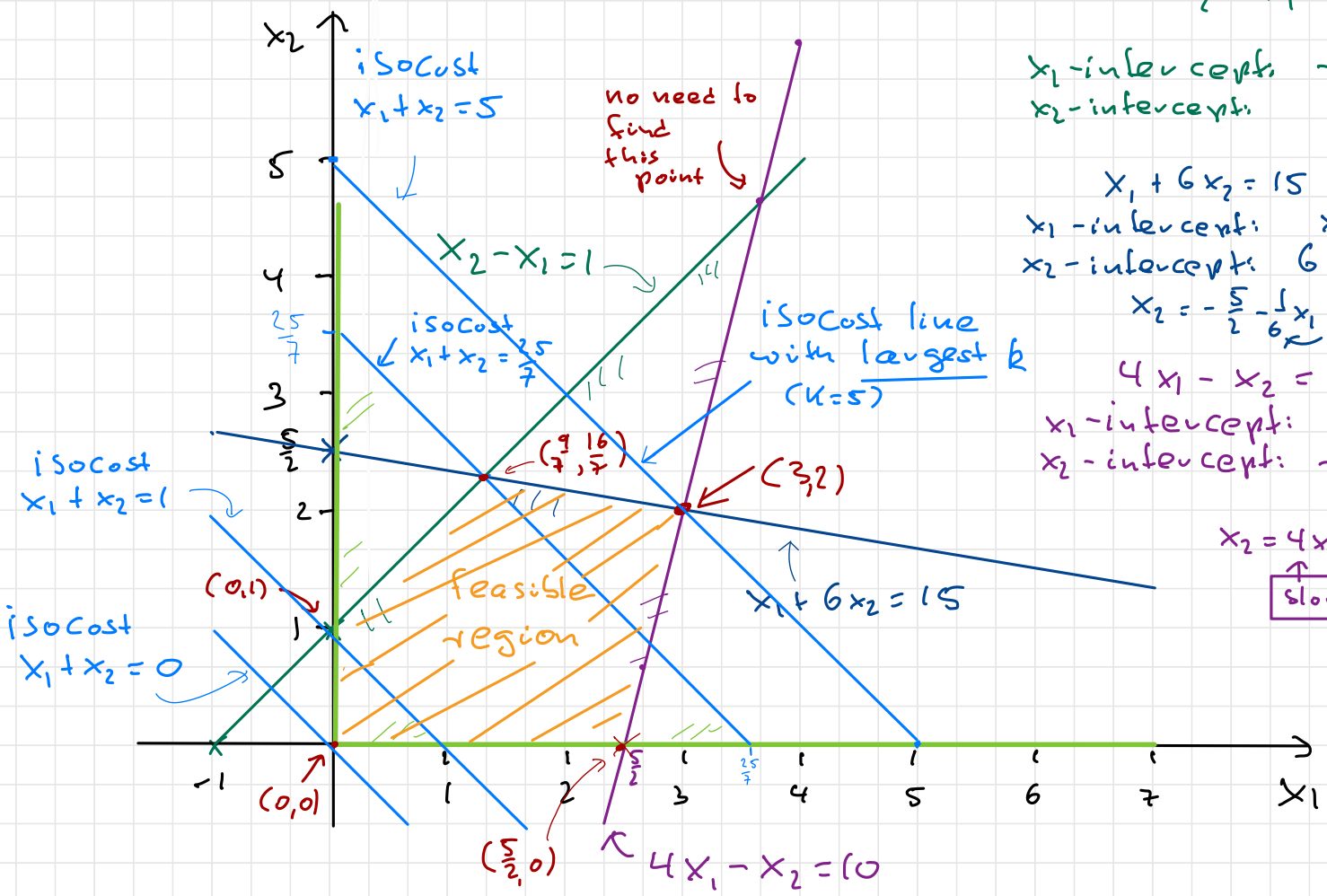
$4x_1 - x_2 = 10 \quad (3)$

(2) + 6(3): $25x_1 = 75$

$x_1 = 3$

$\underset{(3)}{\Rightarrow} x_2 = 4x_1 - 10 = 4 \cdot 3 - 10 = 2$

$(3, 2)$



$x_2 - x_1 = 1$

x_1-intercept: $-x_1 = 1 \Rightarrow x_1 = -1$

x_2-intercept: $x_2 = 1$

$x_1 + 6x_2 = 15$

x_1-intercept: $x_1 = 15$

x_2-intercept: $6x_2 = 15 \Rightarrow x_2 = \frac{5}{2}$

$x_2 = -\frac{5}{2} - \frac{1}{6}x_1$ $\boxed{\text{slope } -\frac{1}{6}}$

$4x_1 - x_2 = 10$

x_1-intercept: $4x_1 = 10$, $x_1 = \frac{5}{2}$

x_2-intercept: $-x_2 = 10$

$\hookrightarrow x_2 = -10$

$x_2 = 4x_1 - 10$

$\boxed{\text{slope } 4}$

Method 1. Sketch the isocost lines

$$f(x_1, x_2) = x_1 + x_2 = k$$

and find the largest possible k. [Maximize!!]

$$[\, x_1 + x_2 = k \Rightarrow x_2 = k - x_1$$

x_2-intercept k slope -1

From the sketch we see: $k = 5$ is largest.

The line $x_1 + x_2 = 5$ touches the feasible region at the point $(x_1, x_2) = (3, 2)$

$\Rightarrow f$ is largest when $x_1 = 3, x_2 = 2 \Rightarrow f(3,2) = \underline{\underline{5}}$

Method 2. Plug the 5 "corner points" into the objective function $f(x_1, x_2) = x_1 + x_2$.

$(0,0)$: $0+0=0 \Leftarrow$ Min.

$\left(\frac{5}{2},0\right)$: $\frac{5}{2}+0=2.5$

$(3,2)$: $3+2=5 \Leftarrow$ Max

$\left(\frac{9}{7},\frac{16}{7}\right)$: $\frac{9}{7}+\frac{16}{7}=\frac{25}{7}\approx 3.57$

$(0,1)$: $0+1=1$

Answer: f is largest at $x_1=3$, $x_2=2$.

$f(3,2)=5$

Example LP5: Modify Exercise 1 to:

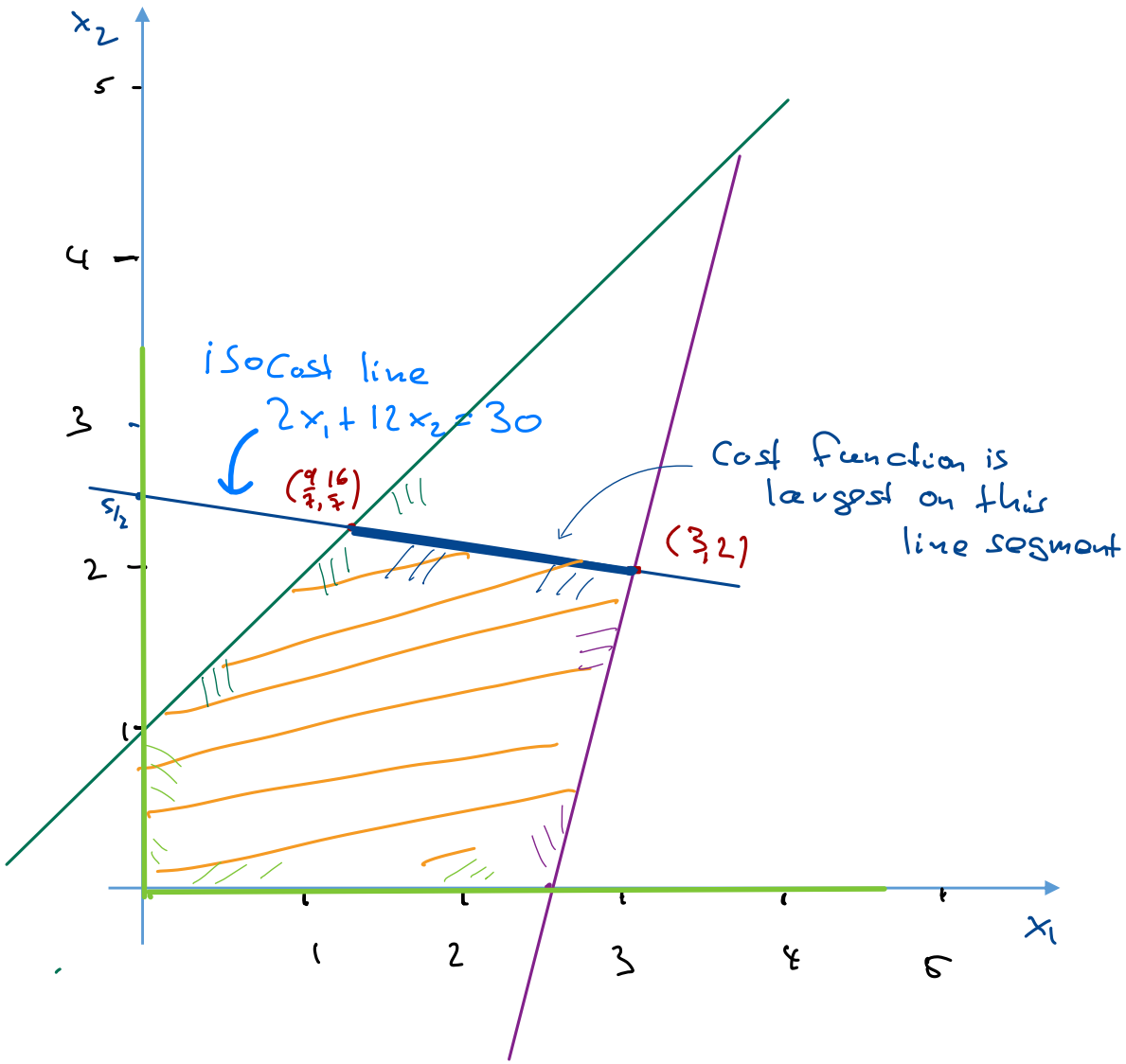
maximize $f(x_1, x_2) = 2x_1 + 12x_2$

subject to:

$$\begin{cases} x_2 - x_1 \le 1 \\ x_1 + 6x_2 \le 15 \\ 4x_1 - x_2 \le 10 \\ x_1, x_2 \ge 0 \end{cases}$$

Same feasible region as in Exercise 1.

We see that every point on the thick blue line segment is an optimal solution.



Example: Modify Exercise 1 to:

maximize $f(x_1, x_2) = x_1 + x_2$

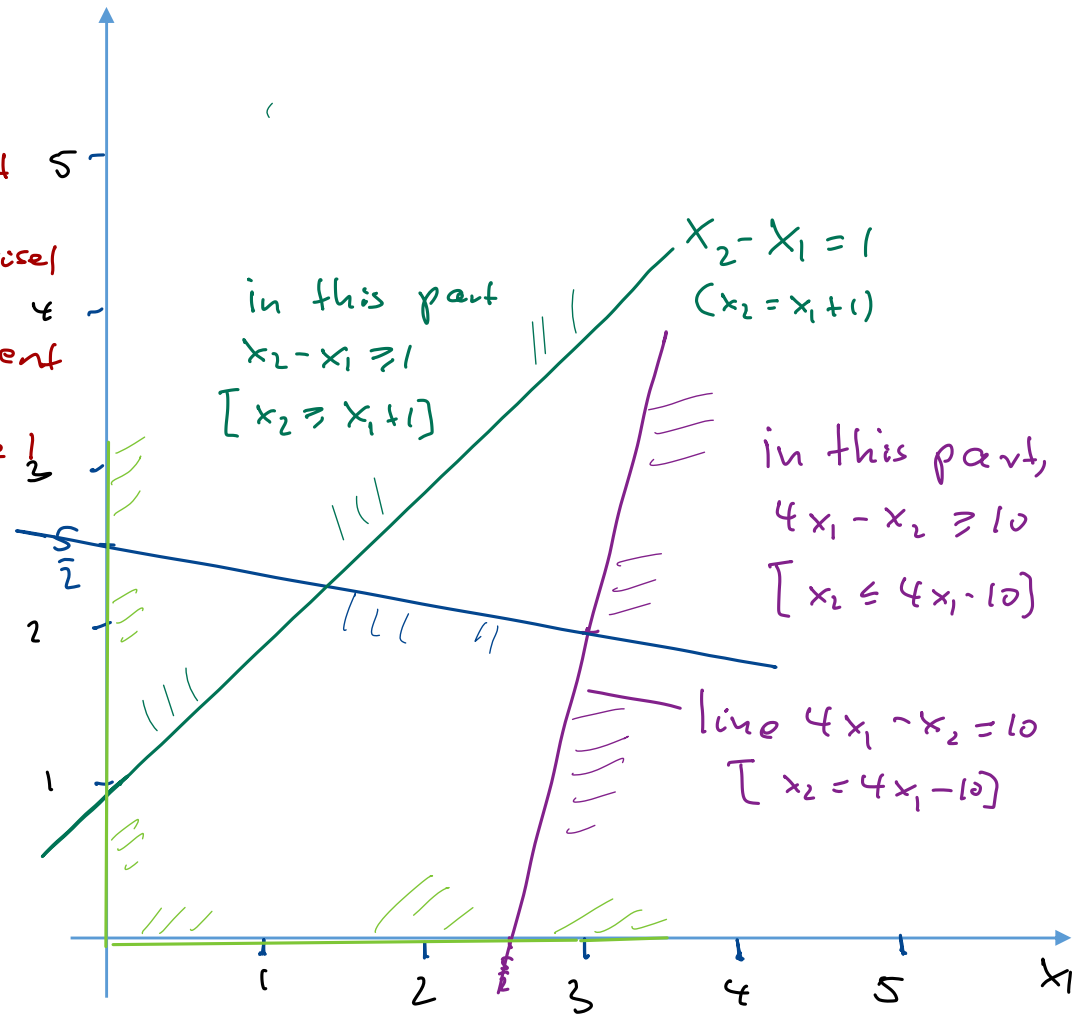
subject to

$$x_2 - x_1 \geq 1$$ different from Exercise 1

$$x_1 + 6x_2 \leq 15$$

$$4x_1 - x_2 \geq 10$$ different from Exercise 1

$$x_1,\ x_2 \geq 0$$



We see that there is NO feasible point (x_1, x_2).

The feasible region Ω is empty: This is called an *infeasible* problem.